Fecal Immunochemical Testing for Colorectal Cancer Screening Outside High-income Countries

A Systematic Review and Meta-Analysis

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Goal: Investigate fecal immunochemical testing (FIT) for CRC screening outside high-income countries (HIC).

Background: Colorectal cancer (CRC) incidence in low- and middle-income countries (LMIC) is rising, but optimal screening strategies in resource-limited settings remain unclear. FIT is an evidence-based, low-cost modality for population-based CRC screening. We performed a systematic review with meta-analysis to better understand the diagnostic performance of FIT for detecting CRC among average-risk individuals living outside HICs.

Study: A systematic search of studies conducted in LMICs, defined by the World Bank, was conducted on August 8, 2024. Studies were eligible if they assessed FIT as a diagnostic test for CRC in asymptomatic, average-risk participants. Pooled estimates for FIT sensitivity and specificity were calculated using a bivariate Bayesian statistical analysis (MetaBayesDTA).

Results: Our search returned 333 distinct studies. Of these, 263 were excluded after title and abstract screening. We reviewed 61 full texts with 7 meeting inclusion/exclusion criteria. These studies were published between 2015 and 2022 and included 100,619 participants from Argentina, Brazil, China, Iran, and Thailand. Most used a FIT hemoglobin cutoff of 50 ng/mL (n = 4). FIT return rate ranged from 78% to 99%, and positivity ranged from 2.1% to 29%. FIT sensitivity ranged from 55% to 92%, specificity from 71% to 94%, PPV from 2% to 16%, and NPV from 98% to 100%. After meta-analysis, pooled estimates for sensitivity and specificity for CRC were 75% (95% CI: 64%-85%) and 89% (95% CI: 77%-95%), respectively.

Conclusions: FIT-based CRC screening shows promise, but data was geographically limited to middle-income countries. This highlights the need for research evaluating screening modalities and completion of the screening continuum in LMICs.

Key Words: colorectal cancer, cancer screening, fecal immunochemical test, low- and middle-income countries (LMICs)

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Worldwide, colorectal cancer (CRC) remains the third most commonly diagnosed cancer and the second leading cause of cancer-related mortality, contributing to ~1.9 million new cancer cases and 935,000 deaths annually.[1–3] Global projections estimate that these numbers will continue to increase as we approach 2040, with the greatest increases occurring in low- and middle-income countries (LMICs).[4] This rising burden of CRC in LMICs is thought to be driven by several factors, including demographic shifts, dietary changes, urbanization/increasingly sedentary lifestyles, and limited access to early detection and treatment.[2]

CRC is well-suited for screening due to the rising burden of incidence and mortality, improved outcomes with early detection, and the availability of evidence-based tests.[4–9] However, one of the greatest barriers to CRC screening in many LMICs, specifically, is the lack of robust endoscopic capacity.[10,11] Therefore, initial CRC screening with stool-based tests is an attractive strategy. Randomized controlled trials of fecal occult blood tests (FOBT) have demonstrated a 15% to 33% decrease in CRC mortality rates, highlighting stool-based testing as a viable screening strategy in high-income countries (HICs).[5–8] Since those studies on FOBT were published, fecal immunochemical tests (FITs) have largely replaced FOBT as the low-cost stool-based screening modality in HICs. Unlike FOBT, which cannot distinguish between human and nonhuman blood, FIT uses a specific antibody for human hemoglobin to detect small amounts of blood in stool. FITs are noninvasive tests that are widely used for population-level CRC screening among average-risk patients in HICs due to their ease of use, low cost, acceptable diagnostic accuracy, requirement of only 1 or 2 stool samples, and no need for dietary restrictions before testing (unlike FOBT). Furthermore, screening for CRC with FIT has been

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C.P.C. and L.R.: conceptualized and designed the study. S.C.: edited the protocol and developed the methods for the systematic search, developing the terms with C.P.C. S.C.: performed the literature search and set up the Covidence platform. C.P.C. and L.R.: performed the study screening, data extraction, and quality assessments. C.P.C.: performed the statistical analyses and prepared the tables and figures. S.C., B.A.S., and N.M.T.: provided critical input on data interpretation, manuscript preparation, as well as clinical and global health implications. C.P.C. and L.R.: drafted the manuscript. S.C., B.A.S., and N.M.T.: supervised the project and critically reviewed the manuscript. All authors have reviewed the final manuscript.

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shown to reduce CRC mortality, as well as disparities in screening.[9,12]

Despite the increasing burden of CRC in LMICs and the availability of low-cost, stool-based screening tests, many LMICs do not have organized screening programs due to limited health care infrastructure, financial constraints, and a strained workforce with a relative lack of specialists (eg, gastroenterologists, surgeons, radiation, and medical oncologists). Early detection is also hindered by limited public awareness of CRC and the concept of screening for the disease before it becomes symptomatic.[13–15] As governments in LMICs develop national CRC screening programs, understanding the diagnostic performance of various screening strategies is crucial for shaping public policy expectations, assessing cost-effectiveness, and guiding the development of infrastructure to support the full screening pathway.

To date, systematic reviews and meta-analyses of the diagnostic accuracy of FITs have primarily focused on HIC,[16] leaving a gap in the literature regarding its applicability in LMICs. To address this gap, we conducted a systematic review with meta-analysis to evaluate the diagnostic performance of FIT for CRC screening among asymptomatic, average-risk individuals living in LMICs.

MATERIALS AND METHODS

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines in the preparation of this manuscript.[17] The protocol for this study was registered in the National Institute for Health Research PROSPERO (ID: CRD42023455594) registry.

Search Strategy

We conducted a comprehensive search of MEDLINE (via Ovid), Embase (via Elsevier), Web of Science—SCI Expanded, SSCI, and ESCI (via Clarivate), Global Health (via EBSCO), and Global Index Medicus (via WHO) from database inception to August 8, 2024. The search strategy was developed and conducted by a medical librarian with input on keywords from the other authors and peer-reviewed using a modified PRESS checklist by an independent librarian expert to ensure accuracy.[18]

The search terms included a combination of database-specific controlled vocabulary terms and keywords searched in the titles or abstracts for variations of colorectal cancer, fecal immunochemical test (FIT), sensitivity and specificity, and low- and middle-income countries (LMICs) as defined by the World Bank. No restrictions were placed on language or date; editorials, comments, and conference abstracts were excluded. The full search strategies for all databases are provided in Supplemental Table 1, Supplemental Digital Content 1, http://links.lww.com/JCG/B239. All citations were imported into Covidence, a systematic review screening software.[19]

Selection Criteria

We included studies that met the following criteria: (1) evaluated performance characteristics of FIT for early CRC detection; (2) study population living within an LMIC, as defined by the World Bank; (3) study population was asymptomatic and at average risk for CRC; (4) used colonoscopy as the reference standard; and (5) reported results that allowed for reconstruction of the diagnostic accuracy 2×2 table. Studies were excluded if: (1) a review of previously conducted studies (ie, not original research); (2) did not include evaluation of FIT for CRC screening (ie, used FIT for a purpose other than CRC screening); (3) FIT test was not followed-up by colonoscopy; (4) used population-specific risk stratification tools together with FIT to improve diagnostic accuracy of the test (ie, not assessing FIT alone); (5) assessed performance among above average risk populations; and/or (6) results were not available in English language. We also excluded case-control studies that obtained FIT results from patients with known CRCs. We did not have a specific hemoglobin cutoff as part of the inclusion/exclusion criteria. However, if a study reported diagnostic performance at various hemoglobin cutoffs, we chose the cutoff of 100 ng/mL, a commonly used cutoff in HICs.[16,20,21] Two reviewers (C.P.C. and L.R.) independently screened titles and abstracts, followed by full-text review to determine eligibility. Any discrepancies at either phase of screening were resolved by consensus discussion.

Data Extraction

Data were extracted into a standardized form, capturing study characteristics (title, publication year, country, funding source, timeframe, target age, FIT brand, hemoglobin cutoff, FIT completion and positivity rates, colonoscopy completion rates, number of prevalent CRCs, and diagnostic performance metrics). This extraction was performed independently by 2 authors (C.P.C. and L.R.) who cross-checked extracted data to ensure accuracy. The index test was FIT for CRC, the reference standard was colonoscopy, and the primary outcome measures were FIT sensitivity and specificity. When possible, raw data were extracted to construct the diagnostic 2×2 tables. However, when only computed proportions were reported, the 2×2 table was reconstructed based on the reported proportions. If necessary, the FIT cutoff was converted to ng/mL to ensure the same units were being used across studies.

Quality Assessment

Study quality was assessed using the QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies-2) tool.[22] Four domains (patient selection, index test, reference standard, and flow and timing) were evaluated for risk of bias and applicability concerns. Two reviewers (C.P.C. and L.R.) independently assessed quality, with conflicts resolved by consensus discussion. Results of the quality assessment are summarized in Supplemental Figure 1, http://links.lww.com/JCG/B239, and Table 2, Supplemental Digital Content 1, http://links.lww.com/JCG/B239.

Data Synthesis

After data extraction, we constructed 2×2 tables containing the true positive (TP), true negative (TN), false positive (FP), and false negative (FN) results for each study. These were used to determine study-level sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), positive likelihood ratios (PLR), and negative likelihood ratios (NLR). Ninety-five percent confidence intervals (CIs) for proportions (eg, sensitivity, specificity) were calculated using the Wilson score method. CIs for likelihood ratios were estimated using a log transformation approach.

For the meta-analysis, a Bayesian hierarchical model was used to estimate pooled sensitivity and specificity, accounting for small sample size and between-study heterogeneity. This was performed using the MetaBayesDTA application with the priors set to default, which allowed for robust Bayesian meta-analysis of diagnostic test accuracy data.[23] Using this tool, we calculated pooled estimates, as well as a summary receiver operating characteristic (sROC) curve

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to assess the overall diagnostic accuracy of FIT across the included studies. Given our use of a Bayesian hierarchical model, heterogeneity was assessed by the between-study standard deviation (SD) for the logit-transformed sensitivity and specificity. Study-level diagnostic performance measures were calculated in R version 4.4.2, and pooled estimates for sensitivity and specificity were calculated using the MetaBayesDTA too.[23,24] The statistical methods used in this study were discussed with a biostatistician from the Duke Global Health Institute's Biostatistics Core.

Sensitivity Analysis

To assess the robustness of our findings, we conducted a sensitivity analysis by stratifying results by FIT hemoglobin cutoff (50 vs. 100 ng/mL and/or no cutoff reported) to look for threshold bias. We also conducted an analysis excluding Ye and colleagues to see if the large difference in sample size or use of a different reference standard (cancer registry follow-up instead of colonoscopy) affected the results. We also performed an analysis excluding studies that were determined to have high or unclear risk of bias/applicability after QUADAS-2 assessment. Lastly, we conducted sensitivity analyses removing each of the 7 studies one at a time to see if there were any studies that had a particular influence on the pooled estimates.

RESULTS

Study Selection

A total of 681 studies were identified during the systematic search process. After the removal of 348 duplicates, 333 studies were screened. After review of titles and abstracts, 263 studies were excluded, leaving 61 titles for full text review. Of the 61 full texts reviewed, 54 were excluded for the following reasons: wrong setting [eg, HIC (n = 1)], wrong study design (n = 2), wrong patient population [(eg, screening above average risk patients) n = 13], not available in English language (n = 4), and/or did not report enough data to reconstruct the diagnostic 2×2 table (n = 34). Ultimately, 7 studies were included in the meta-analysis.[25–31] This study selection process is detailed in the PRISMA flow diagram shown in Figure 1.

Study Characteristics

The 7 included studies were conducted in middle-income countries (MICs) from 2015 to 2019: Argentina (n = 1), Brazil (n = 1), China (n = 1), Iran (n = 1), and Thailand (n = 3). No studies from low-income countries (LICs) met the inclusion/exclusion criteria. Sample sizes ranged from 156 to 96,043, representing a total of 100,619 participants. Study design varied between a cross-sectional design (n = 2), a prospective cohort design (n = 2), and an unclear design (n = 3). Regarding funding source, 3 studies were funded by government grants, while 2 were funded by university grants (2 studies did not report funding source). Two studies did not explicitly state where participants were recruited, but 3 recruited out of hospitals, while 1 study recruited from a GI clinic, and 1 recruited from the community.

The mean age ranged from 56.9 to 60.6 years, and the proportion of female participants ranged from 34.0% to 69.4%. Most studies (n = 6) reported the brand of FIT used and the hemoglobin cutoff for a positive test (n = 4 used 50 ng/mL, n = 2 used 100 ng/mL cutoff, and n = 1 did not report the cutoff used). FIT completion rates ranged from 78.0% to 99.4%, with subsequent positivity rates ranging from 2.1% to 29.1%. One study used 2 stool samples, while the remaining 6 used a single stool sample.

Completion rates for colonoscopy ranged from 33.9% to 99.6% (note that 3 studies did not report colonoscopy completion rate). Endoscopists were blinded to FIT results in 4 studies (in 3 studies, it was not clear if endoscopists were blinded). The number of prevalent cancers detected during the screening process ranged from 253/96,043 (0.3%) to 9/156 (5.7%). The individual study characteristics are provided in Table 1.

Study Quality and Publication Bias

Assessing the risk of bias using the QUADAS-2 tool, most studies demonstrated a low risk of bias in applicability of the index test (FIT) and reference standard (colonoscopy) domains. However, several studies had unclear bias risks due to insufficient reporting on patient flow and timing, as well as patient selection (eg, not entirely clear what the target population was and how average risk was defined). The risk of bias assessment summary is shown in Supplemental Figure 1, Supplemental Digital Content 1, http://links.lww.com/JCG/B239 and Table 2, http://links.lww.com/JCG/B239

To assess for publication bias, a funnel plot was created (Supplemental Figure 2, Supplemental Digital Content 1, http://links.lww.com/JCG/B239) with the *x*-axis representing the effect size in terms of the diagnostic odds ratio (DOR), which is a composite measure combining sensitivity and specificity, and the *y*-axis representing the precision of the study. This plot showed some asymmetry, with more studies to the right of the overall DOR (indicating that higher DORs may be overrepresented in the literature) and potential small study effects (higher DOR reported by smaller studies). Both findings indicate that there could be publication bias occurring.

Diagnostic Performance of FIT for CRC Detection Across Individual Studies

The sensitivity of FIT for detecting CRC ranged from 0.56 (95% CI: 0.34-0.75)[26] to 0.92 (95% CI: 0.65-0.99),[29] while the specificity ranged from 0.71 (95% CI: 0.68-0.74)[25] to 0.99 (95% CI: 0.95-1.00).[31] There was a wide range of DORs, from 14.5 (95% CI: 1.5-142.9)[30] to 145.0 (95% CI: 20.3-1036.3).[31] Predictive values also showed substantial variation across studies, with PPV ranging from 0.02 (95% CI: 0.01-0.05)[25] to 0.4 (95% CI: 0.41-0.93)[31] and NPV ranging from 0.980 (95% CI: 0.942-0.993)[31] to 0.999 (95% CI: 0.999-0.999).[28] PLRs ranged from 3.0 (95% CI: 2.2-4.1)[25] to 14.7 (95% CI: 9.0-24.0)[26] and NLRs ranged from 0.11 (95% CI: 0.016-0.68)[29] to 0.46 (95% CI: 0.28-0.77).[26] Table 2 summarizes the diagnostic performance metrics for each study.

Pooled Diagnostic Performance of FIT for CRC Detection

Forest plots showing individual and pooled estimates for sensitivity and specificity of FIT for detecting CRC in average-risk participants are shown in Figure 2. After meta-analysis, the pooled sensitivity was 0.75 (95% CI: 0.64-0.85) and the pooled specificity was 0.89 (95% CI: 0.77-0.95). The sROC is shown in Figure 3.

Heterogeneity Assessment

The between-study standard deviation (SD) for the logit-transformed sensitivity was 0.393 (SD = 0.322), indicating that there is moderate heterogeneity in the sensitivity across

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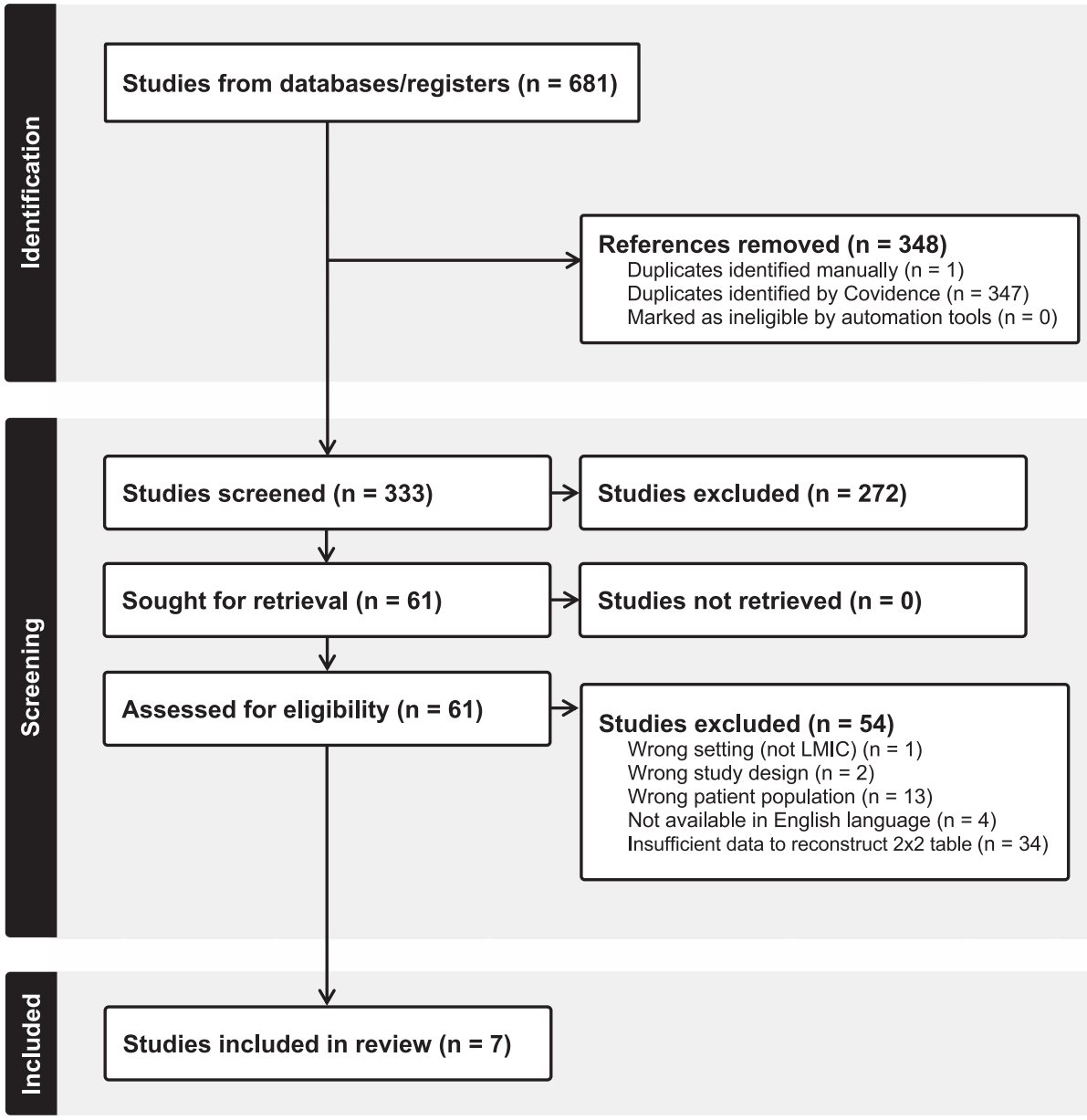


FIGURE 1. Flow diagram summarizing systematic search and study selection.

studies (ie, moderate difference in FIT's ability to detect TPs across studies). The between-study SD for the logit-transformed specificity was substantially larger at 1.062 (SD = 0.316), indicating considerable heterogeneity in specificity across studies (ie, FIT's ability to correctly identify TNs varied across studies). In other words, FIT was more consistent in identifying TPs across studies but showed substantial variability in false positive rates.

Sensitivity Analysis

When excluding studies with a hemoglobin cutoff of 50 ng/mL (Aniwan and colleagues; Siripongpreeda and colleagues; Teixeira and colleagues; Pereyra and colleagues; Dolatkhah and colleagues), the pooled sensitivity and specificity remained similar. The sensitivity, excluding those studies, was 0.75 (95% CI: 0.51-0.89) versus 0.75 (95% CI: 0.64-0.85) in the complete analysis. The specificity, excluding those studies, was higher at 0.91 (95% CI: 0.75-0.95) versus 0.89 (0.95% CI: 0.77-0.95) when including all studies. When excluding studies with a high or unclear applicability and/or risk of bias, the pooled sensitivity decreased to 0.72 (95% CI: 0.53-0.85) from 0.75 (95% CI: 0.64-0.85) and the specificity remained similar at 0.90 (95% CI: 0.70-0.97) from 0.89 (0.95% CI: 0.77-0.95). When excluding the much larger study performed by Ye and colleagues, the 95% CIs widened, but the pooled estimates remained the same as the complete analysis. Lastly, when we sequentially removed each study from the analysis, the sensitivities ranged from 0.74 when Teixeira and colleagues were removed to 0.77 when Siripongpreeda and colleagues were removed. Specificities ranged from 0.87 when Siripongpreeda and colleagues were removed to 0.91 when Aniwan and colleagues were

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TABLE 1. Characteristics of Included Studies.

References	Country	Study design	Funding source	Recruitment site	Endoscopist blinded	Time-frame	Target age	Mean age	Female (%)	FIT brand	No. FIT samples	Hgb cutoff (ng/mL)	Total participants (N)
Aniwan et al[25]	Thailand	Cross-sectional	University hospital grant	Hospital	Yes	2013-2014	50-75	60.6	65.0%	SD Bioline FOB (Korea)	1	50	948
Siripongpreeda et al[26]	Thailand	NR	University hospital grant	NR	NR	2009-2010	50-65	56.9	69.4%	Abon Biopharm Hangzhou (China)	1	50	1404
Aniwan et al[27]	Thailand	Cross-sectional	Government grant	Hospital	Yes	2014-2016	50-75	60.4	61.7%	OC-SENSOR (Japan)	1	100	1479
Ye et al[28]	China	Cohort	NR	Community	NR	2007-2014	40-74	NR	NR	Wanhua Biotech Co. Ltd (China)	2	100	96043
Teixeira et al[29]	Brazil	NR	Government grant	GI clinics	Yes	2015-2016	50-75	60.5	64.9%	OC-Light i-FOBT (Japan)	1	50	316
Pereyra et al[30]	Argentina	Cohort	NR	Hospital	Yes	2015-2017	50-75	57.9	46.2%	MONTEBIO FOB (Argentina)	1	50	273
Dolatkhah et al[31]	Iran	NR	Government grant	NR	NR	2016-2019	≥ 50	NR (77.0% ≥ 50)	34%	NR	1	NR	156

References	FIT completion (%)	No. CRCs (prevalence)	FIT positivity (%)	Colonoscopy completion (%)
Aniwan et al[25]	99.4%	7/948 (0.7%)	29.1%	99.1%
Siripongpreeda et al[26]	97.2%	18/1404 (1.3%)	4.4%	99.6%
Aniwan et al[27]	99.2%	14/1479 (0.9%)	7.3%	98.6%
Ye et al[28]	85.0%	12/316 (3.8%)	19.9%	33.9%
Teixeira et al[29]	91.2%	253/96043 (0.3%)	23.1%	NR
Pereyra et al[30]	91.0%	4/273 (1.5%)	2.1%	NR
Dolatkhah et al[31]	78.0%	9/156 (5.7%)	5.3%	NR

NR indicates not reported.

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TABLE 2. Diagnostic Performance of Fecal Immunochemical Tests for Detecting Colorectal Cancer

References	Sensitivity* (95% CI)	Specificity* (95% CI)	DOR† (95% CI)	PPV* (95% CI)	NPV* (95% CI)	Positive LR†‡ (95% CI)	Negative LR† (95% CI)
Aniwan et al[25]	0.86 (0.49-0.97)	0.71 (0.68-0.74)	14.9 (1.8-124.4)	0.02 (0.01-0.05)	0.999 (0.992-1.000)	3.0 (2.2-4.1)	0.20 (0.03-1.23)
Siripongpreeda et al[26]	0.56 (0.34-0.75)	0.96 (0.95-0.97)	31.7 (12.0-83.8)	0.16 (0.09-0.28)	0.994 (0.988-0.997)	14.7 (9.0-24.0)	0.46 (0.28-0.77)
Aniwan et al[27]	0.79 (0.52-0.92)	0.93 (0.92-0.95)	51.7 (14.1-188.4)	0.10 (0.06-0.17)	0.998 (0.994-0.999)	11.9 (8.5-16.6)	0.23 (0.084-0.63)
Ye et al[28]	0.76 (0.70-0.80)	0.90 (0.90-0.91)	28.9 (21.7-38.5)	0.02 (0.018-0.023)	0.999 (0.999-0.999)	7.8 (7.3-8.4)	0.27 (0.22-0.34)
Teixeira et al[29]	0.92 (0.65-0.99)	0.80 (0.75-0.84)	42.9 (5.4-338.9)	0.15 (0.09-0.25)	0.996 (0.977-1.00)	4.5 (3.4-5.9)	0.11 (0.016-0.68)
Pereyra et al[30]	0.75 (0.30-0.95)	0.83 (0.78-0.87)	14.5 (1.5-142.9)	0.06 (0.02-0.17)	0.996 (0.975-0.999)	4.4 (2.4-8.2)	0.30 (0.055-1.65)
Dolatkhah et al[31]	0.67 (0.35-0.88)	0.99 (0.95-1.0)	145.0 (20.3-1036.3)	0.40 (0.41-0.93)	0.980 (0.942-0.993)	49.0 (11.5-209.3)	0.34 (0.13-0.85)

*95% CI was calculated using the Wilson Score method with "prop.test" in R version 4.4.2.
†Calculated using the "madad" function in the R package "mada."
‡Correlation of sensitivities with false positive rates, rho = 0.770 (95% CI: 0.040-0.964).

removed. The results from these subgroup analyses can be found in Supplemental Table 3, Supplemental Digital Content 1, http://links.lww.com/JCG/B239.

DISCUSSION

Summary of Evidence

In this systematic review with meta-analysis, we sought to evaluate the diagnostic performance of FIT for the detection of CRC in asymptomatic, average-risk people living outside HICs. Our search ultimately yielded 7 studies from MICs with a pooled sensitivity of 0.75 (95% CI: 0.64-0.85) and a pooled specificity of 0.89 (95% CI: 0.77-0.95). These results are comparable to meta-analyses conducted in HICs, where pooled sensitivities range from 0.73 to 0.93 and specificities from 0.90 to 0.94, suggesting that the diagnostic performance of FIT may be robust across diverse patient populations.[16,20,21,32] Our pooled results meet the minimum threshold established by CMS in the United States for CRC screening tests of 0.74 for sensitivity, but fall just short of the 0.90 minimum specificity.[33] Of note, in sensitivity analysis using the 100 ng/mL cut-off, a commonly used cutoff in HICs, specificity met the CMS benchmark. This may be particularly relevant for resource-limited settings, where minimizing false positives may be a priority for efficient allocation of follow-up colonoscopy.

Though a small difference, a lower specificity may not be acceptable in lower resource settings, where a larger proportion of false positives could not be followed up due to a lack of sufficient colonoscopy resources. There have been efforts to tailor strategies using risk stratification with tools like the Asia-Pacific Colorectal Screening (APCS) Score to enhance the efficiency of screening with FIT.[34] These tools incorporate demographic, clinical, and lifestyle factors to categorize individuals into risk groups with the goal of optimizing CRC screening. While care must be taken to ensure the quality and external validity of risk prediction models, pairing risk stratification with FIT testing has the potential to improve diagnostic accuracy.[35,36]

A critical result of this review is that all included studies were conducted in MICs, with no studies meeting inclusion criteria from LICs. Given the epidemiological relevance of CRC to the disease burden in LICs, this is a significant gap in knowledge that merits urgent research efforts to inform global CRC screening guidelines and improve equitable access to early detection strategies. While we found no studies in LICs that met our inclusion/exclusion criteria, foundational work is underway to evaluate the feasibility of FIT in LICs where unique challenges such as higher ambient temperatures and endemic parasitic infections may impact results.[37,38] One such group, Knapp and colleagues, has demonstrated the feasibility and acceptability of FIT among average-risk populations in Nigeria (the first of such studies in sub-Saharan Africa).[39–41] This work lays the groundwork for studies that examine a more complete picture of the diagnostic performance of FIT in resource-limited settings.

In addition, while gross domestic product per capita is commonly used to classify countries' economic status, it may not fully capture the complex determinants of health relevant to CRC screening. The Socio-Demographic Index (SDI), which integrates income, education, and fertility rates, could be an alternative framework for assessing screening feasibility and outcomes across different contexts.

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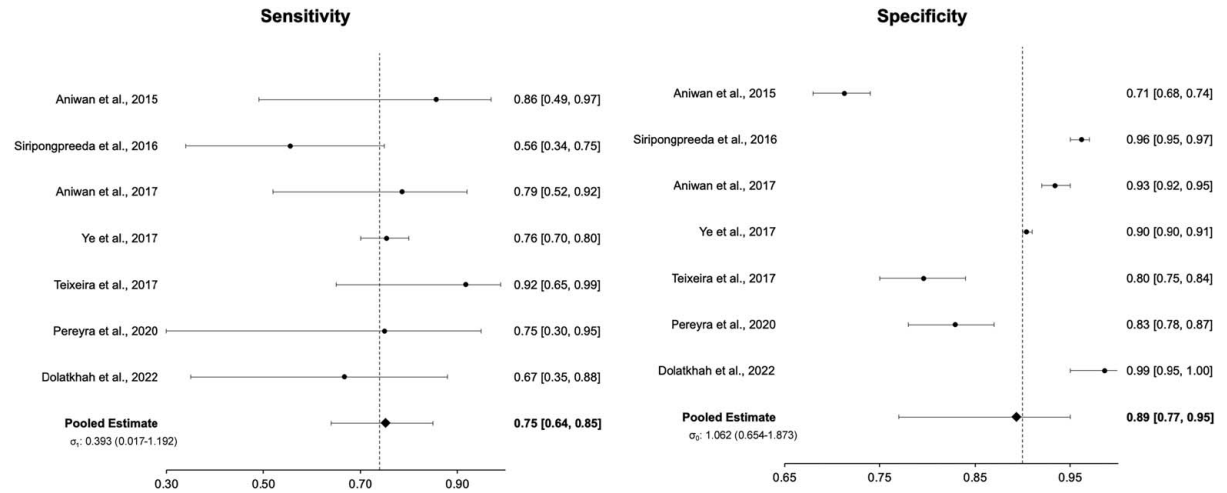


FIGURE 2. Forest plots with pooled sensitivity and specificity for fecal immunochemical tests for the detection of colorectal cancer for all included studies. Vertical dotted lines indicate the Centers for Medicare & Medicaid Services (CMS) minimum thresholds for colorectal cancer screening tests in the United States (sensitivity of at least 0.74 and specificity of at least 0.90).

While GDP per capita is traditionally used to define LMICs, future studies could also consider using SDI to explore FIT performance in various socio-demographic areas rather than the World Bank classification.

The efficacy of FIT-based CRC screening is best realized within organized, population-level screening programs, as seen in many European countries.[42] This contrasts with opportunistic screening, which relies on individual awareness about screening, health care access, and provider recommendations. Successful population-based CRC screening programs require the availability of an effective screening test like FIT integrated with public engagement, test distribution, results tracking, diagnostic follow-up, and surveillance. Future studies should focus on evaluating FIT in a broader range of LMICs, standardizing test protocols, comparing qualitative versus quantitative FIT, determining ideal thresholds for various settings, the benefit of serial testing to reduce false-positive rates, cost-effectiveness analyses between CRC screening modalities, cultural acceptance of stool-based testing, and performance for detecting precancerous lesions.

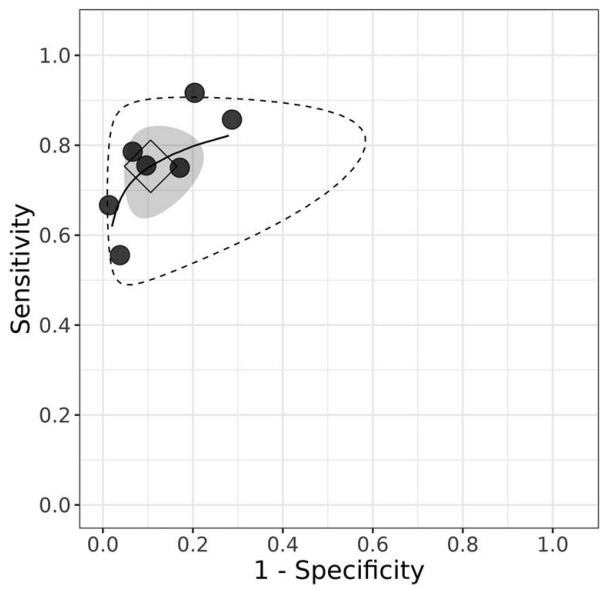


FIGURE 3. Summary receiver operator curve (solid black line) with black dots representing individual studies. The dotted line represents the 95% prediction region calculated from the bivariate model. The gray shaded area represents the 95% credible region from the bivariate model.

While FIT shows promise as a CRC screening modality outside HICs, implementation of a successful CRC screening program extends beyond diagnostic test performance alone. Barriers to the adoption and implementation of FIT must be studied in parallel with the entire screening continuum to ensure that programs are feasible and effective within the intended context. In addition to evaluating the performance of stool-based screening modalities, research must also address strategies for identifying the appropriate screening cohort, implementing screening tests longitudinally (including distribution, completion, return, and processing over time), communicating results, ensuring appropriate follow-up for positive and negative tests, and linking those diagnosed with CRC to treatment. FIT implementation studies from LMICs are generally hopeful, with studies from Nigeria, Brazil, Malaysia, and Mexico showing general acceptability of FIT among patients and completion rates comparable to HICs.[38,40,41,43–46] However, these studies were small and called for further study of large-scale programmatic FIT-based CRC screening implementation. In a scoping review on the implementation of CRC screening in LMICs, Schliemann and colleagues found that some of the greatest challenges for screening programs in LMICs were lack of CRC knowledge among the public, understaffed health workforce with limited availability of qualified endoscopists, health care system infrastructure, upfront costs, and need for health policy development.[47] Advancing research on CRC screening modalities and their implementation is a priority to guide policy and work toward the United Nations Sustainable Development Goal of reducing premature mortality from noncommunicable diseases.[48]

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Limitations

This review has several limitations. First, the small number of studies and their geographic concentration in MICs limit the generalizability of our findings to other MICs and LICs. Methodological differences between studies may also affect the interpretation and generalizability of our results. Of particular importance, there was considerable variability in diagnostic performance between studies, with FIT sensitivity ranging from 0.56 to 0.92 and specificity ranging from 0.71 to 0.99. This notable heterogeneity between the studies included in our review could have been introduced by a host of factors. Variations in gastrointestinal disease patterns and endemic infections, such as parasitic diseases, may influence gastrointestinal bleeding patterns and affect FIT performance.[38] Differences in FIT brand, hemoglobin cutoff, instructions for sample collection, transport (FIT tests are affected by both time and temperature), and laboratory processing could all contribute to variability between studies.[16,37,49–51] Most studies did not report detailed information on sample handling, storage conditions, or time to analysis, limiting our ability to assess these factors. Another important source of variability between studies was the differences in how average risk was defined. If the populations being assessed had different risk profiles, the diagnostic performance would be skewed. Finally, there was also limited information provided in these studies regarding the quality of endoscopies (eg, bowel prep score, endoscopists' adenoma detection rate). It is known that poor bowel prep and endoscopists with low adenoma detection rates can affect the ability to detect cancer at the time of colonoscopy.[52–54] Such a wide degree of variation underscores both the potential benefit of reporting standards for the study of CRC screening modalities, as well as the importance of local validation before implementation.

CONCLUSION

Our findings suggest that performance characteristics of FIT for CRC screening in MICs may be comparable to HICs, potentially indicating feasibility and acceptability as part of population-based CRC screening programs in these settings. Notably, a critical research gap was identified in that no studies from LICs have published the diagnostic performance of FIT as a CRC screening modality. Further, in addition to clarifying the performance characteristics of stool-based screening tests, a better understanding of implementation challenges will be essential to develop comprehensive, evidence-based CRC screening programs worldwide by enabling policymakers to set realistic targets, optimize current resource allocation, and align infrastructure investments to maximize the population-level impact of screening.

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